

Data Quality Report

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Executive Summary

The supplied dataset covers 2024-01-01 00:02:27 to 2026-09-06 21:23:52 across 17 tables. The pipeline preserves raw files, audits keys/FKs/timestamps/categories, and creates clean/golden outputs.

Major recovery impact: raw successful payments totaled INR 1,341,485,926; validated successful payments totaled INR 1,149,682,230; duplicate/payment-status treatment reduced recovery by INR 191,803,696.

Major Findings

Issue	Detection	Treatment	Affected Records	Business Impact
agents duplicates	duplicate audit	classify, do not blanket drop	30000 duplicate-ID rows	affects source trust
borrowers duplicates	duplicate audit	classify, do not blanket drop	28151 duplicate-ID rows	affects source trust
calls duplicates	duplicate audit	classify, do not blanket drop	2700 duplicate-ID rows	affects source trust
whatsapp_events duplicates	duplicate audit	classify, do not blanket drop	1200 duplicate-ID rows	affects source trust
payments duplicates	duplicate audit	classify, do not blanket drop	1000 duplicate-ID rows	affects source trust
campaigns duplicates	duplicate audit	classify, do not blanket drop	0 duplicate-ID rows	affects source trust
daily_targeting duplicates	duplicate audit	classify, do not blanket drop	0 duplicate-ID rows	affects source trust
agent_sessions duplicates	duplicate audit	classify, do not blanket drop	0 duplicate-ID rows	affects source trust
borrowers primary key	key audit	keep raw, resolve in clean/golden	19585 duplicate keys	affects joins
agents primary key	key audit	keep raw, resolve in clean/golden	29000 duplicate keys	affects joins
calls primary key	key audit	keep raw, resolve in clean/golden	1350 duplicate keys	affects joins
whatsapp_events primary key	key audit	keep raw, resolve in clean/golden	600 duplicate keys	affects joins
payments primary key	key audit	keep raw, resolve in clean/golden	500 duplicate keys	affects joins

Foreign Keys

Unmatched FK rows are written to `data/processed/audit\_outputs/fk\_integrity.csv`; they are not silently deleted because they may represent late-arriving or historical source mismatches.

Cleaning Impact

See `data/processed/audit\_outputs/cleaning\_impact.csv`.

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Residual Risks

Client and language dimensions are unavailable in supplied data. Actual cost facts are unavailable, so cost per recovered rupee and investment ROI are modeled assumptions rather than observed financial accounting.